Attention-based Memory Supports Learning of Semantically Rich Representations

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Motivation

Cognitive Science and Neuroscience: Episodic memory (EM)—a mechanism for rapid and durable storage of previous experiences, or episodes—is thought to be a core element underlying learning and generalization in humans. EM can be viewed as a complementary learning system to the longer-term, slower learning of semantic structure in cortex [1,2]. However, could episodic memory play a deeper, more active role in shaping long-term semantic knowledge?

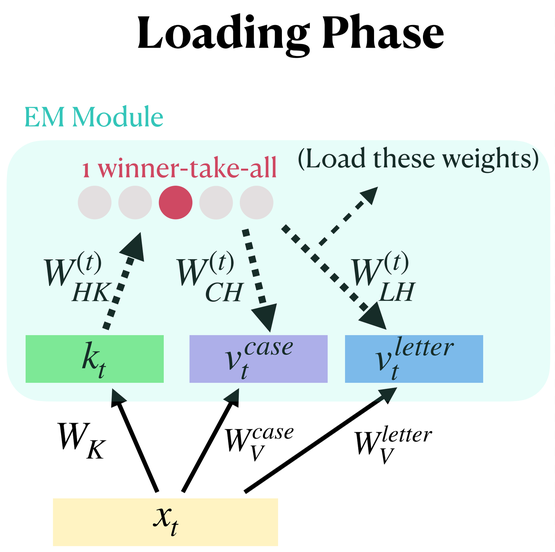
Machine Learning: How does the self-attention mechanism [3] affect feature learning? Given that various task-relevant attributes (along different "dimensions") must be learned in order to solve large, complex tasks, does an attention-based based architecture bias a model towards learning certain feature dimensions more than others?

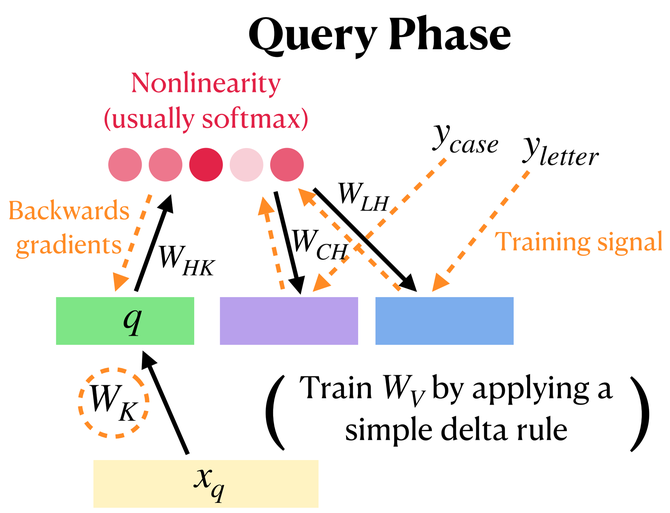
Methods

Task Design and Model Architecture

- We study a simple task that we call the **case-letter task**: a sequence of C letters (with varying cases), each encoded as an $d = 2L$ dimensional one-hot vector, are presented as an "initial exposure"
- These letters are presented a second time, and the goal is to output both the case and letter type of each letter; a weighted MSE loss is used for training
- Note: W_V (lazily) learns by linear regression!

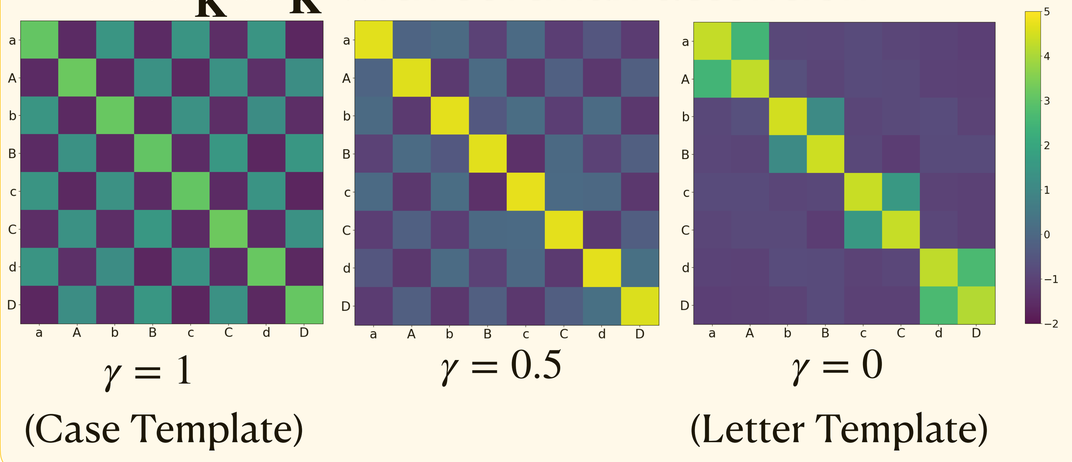
$$(W_K)_{ij} \overset{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{\theta_0}{N}\right) \quad \mathcal{L} = \gamma \mathcal{L}_C + (1-\gamma)\mathcal{L}_L, (0 \le \gamma \le 1)$$





Results

$W_K^T W_K$ with softmax attention



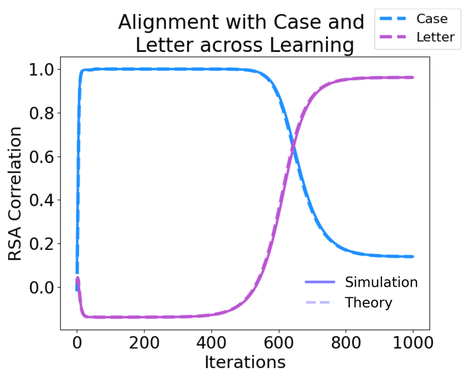
A Mathematical Theory of Linear Attention

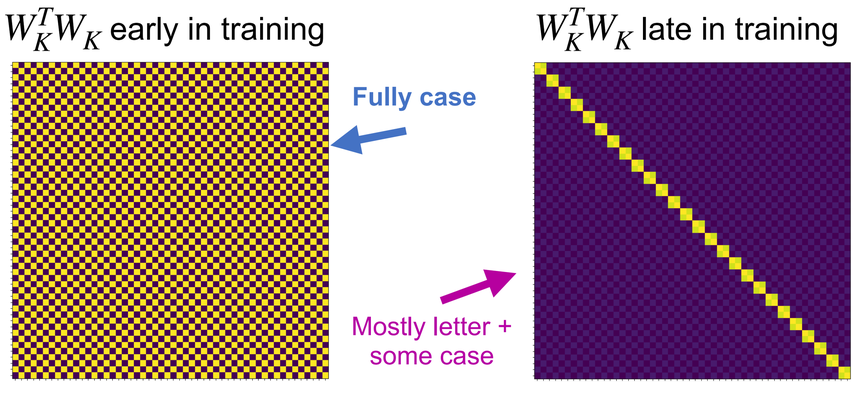
Assumptions: 1. $C = 2L$ 2. $N > d = 2L$ 3. $W_V = W_V^\star$ has finished learning.

Theorem: Define $Q(t) = W_K(t)^T W_K(t)$. Let $Y_C \in \mathbb{R}^{d_C \times n}$, $Y_L \in \mathbb{R}^{d_L \times n}$ be the targets along the two dimensions, and define $Z = \gamma Y_C^T Y_C + (1-\gamma) Y_L^T Y_L$ with $Z = USV^T$ as its SVD. Then, training the model by gradient flow with a fixed learning rate $\eta = \Theta(1)$, the dynamics of Q can be given in closed form:

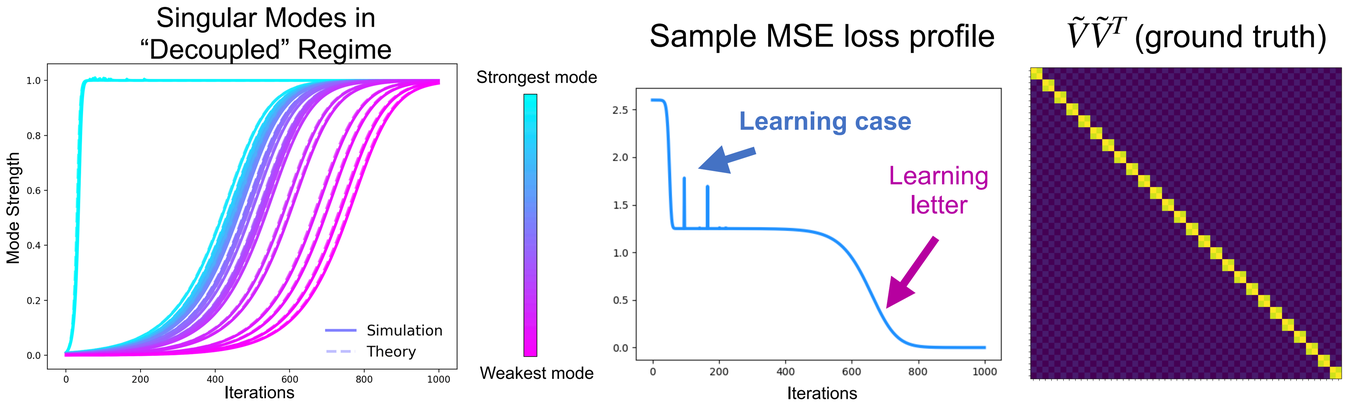
$$Q(t) = V e^{\eta t S} \left(V^T Q_0^{-1} V + e^{2\eta t S} - I_n \right)^{-1} e^{\eta t S} V^T$$

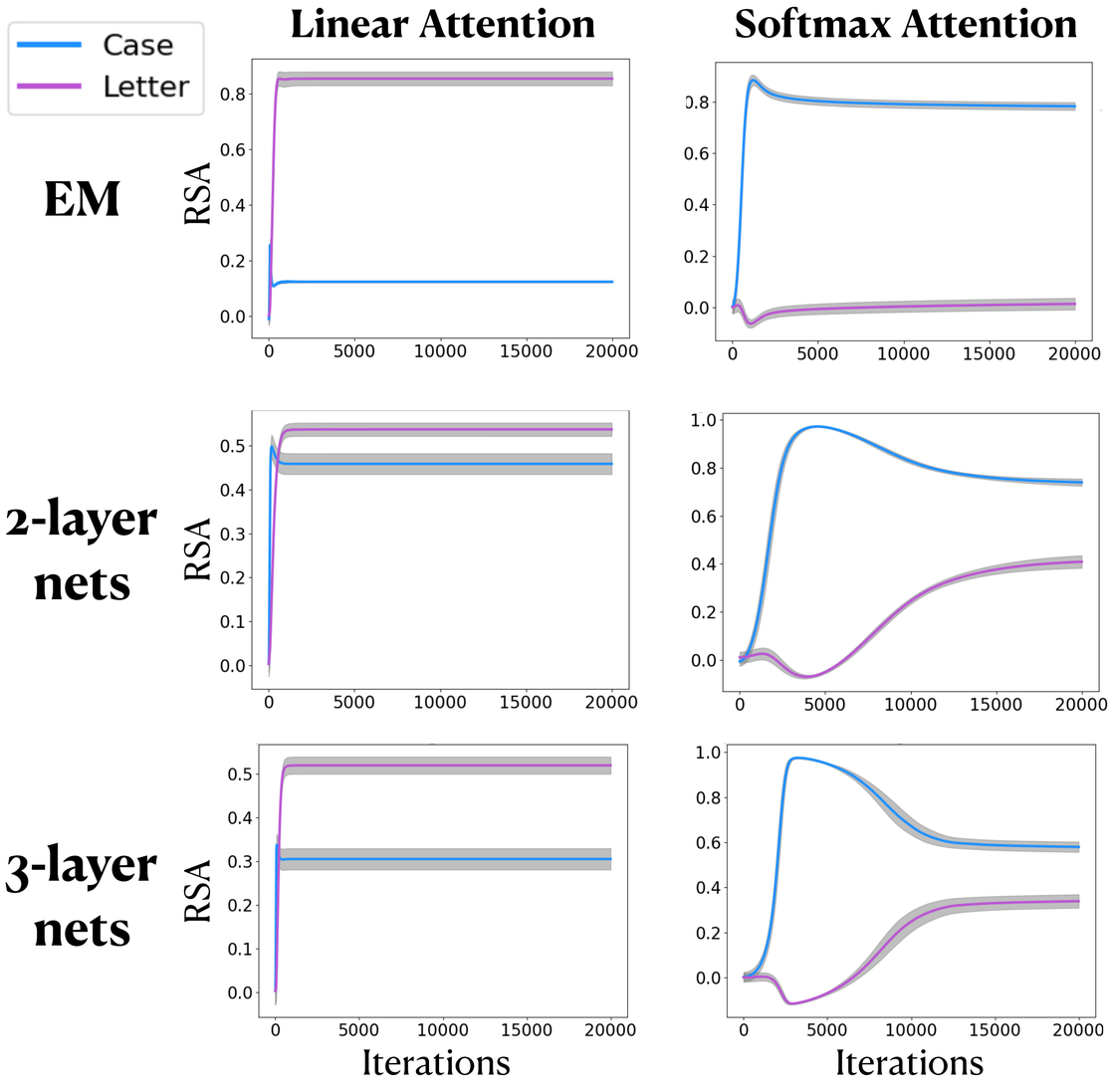
Key Idea: $\frac{dQ}{dt} = \eta \left(ZQ + QZ - 2QZQ \right)$ is a matrix Riccati differential equation [4], and can be solved!





Corollary (informal): For generic initializations $\lim_{t\to\infty} Q(t) = \tilde{V}\tilde{V}^T$, where $\tilde{V}$ is the truncated version of V. Moreover, in a "decoupled" setting [2] where Q_0 has V as its eigenvectors, $Q(t)$ will have V as its eigenvectors for all time and each eigenvalue will follow a sigmoidal trajectory to 1.





For $Q = W_K^T W_K \in \mathbb{R}^{d \times d}$, $Q_A \left(A \in \{C, L\}\right)$ a ground truth template,

$$\text{RSA}(Q, Q_A) = \frac{\langle Q\tilde{I}, Q_A\tilde{I} \rangle}{||Q\tilde{I}||_F ||Q_A\tilde{I}||_F} \text{ where } \tilde{I} = \mathbf{1}_d \mathbf{1}_d^T - I_d .$$

Conclusions

An EM / attention-based architecture is capable of supporting the learning of *compositional* task-relevant representations, but with preferential learning of lower-dimensional (resp. higher-dimensional) features in the softmax (resp. linear) setting; theory for softmax remains to be explored…

References

1. McClelland et al. (1995). *Complementary learning systems in hippocampus & neocortex*. Psychol. Rev.

2. Saxe et al. (2019). *Mathematical theory of semantic development in deep nets*. PNAS

3. Vaswani et al. (2017). *Attention is all you need*. NeurIPS

4. Sasagawa (1982). *Finite escape in matrix Riccati equations*. IEEE TAC